

Gen AI & Agentic Engineering Curriculum

Module 1: Foundations & Setup

Local Environment Setup

- Student Lab Installation (Local)

Hardware Requirements

- GPU, RAM, and Storage specifications for AI/LLM workloads

Introduction to AI

- Data Science (DS) & Machine Learning (ML)
- Large Language Models (LLMs) & Transformers

Computer Science Fundamentals

- Graph Data Structures and Algorithms
- Basic Data Structures & Algorithms (DSA)

Module 2: LLM Engineering & Architecture

Core Engineering

- Prompt Engineering
- Context Engineering
- Token Optimization

LLM Internals

- Model Architectures
- Multimodal capabilities (Chat with File, Image & Video Generation)

High-Level Design (HLD)

- Web HLD Design Thinking for Architects

Module 3: Databases & Knowledge Management

Database Systems

- RDBMS vs. Vector Databases
- OpenSearch & Indexing strategies

Data Processing

- Embeddings
- Vectorless Memory
- Semantic Search

Module 4: Generative & Agentic AI

Core Concepts

- Finding Problems for AI to Solve
- Retrieval-Augmented Generation (RAG)
- Knowledge-Augmented Generation (KAG)
- Hypothetical Document Embeddings (HyDE)

Agent Architecture

- Tool Calling & Agent Loops
- Memory Systems (Short-term & Long-term)
- Talking to Databases (Text-to-SQL)

Integration & Security

- Model Context Protocol (MCP) Integration
- API Integration
- Security Protocols & Audits

Module 5: The Claude Ecosystem

Development Tools

- Claude Code (Co-worker / Coding Agent)
- Claude Design & Artifacts

Automation

- Desktop Automation
- Browser Automation
- Agentic Tools & Skill Design

Advanced Features

- MCP Integration
- Code Reviewer Agents
- Building Knowledge Graphs for Memory

Module 6: Proprietary Models & Selection Strategy

Model Landscape

- OpenAI, Perplexity, Google Gemini
- NotebookLM, Kimi, Minimax, GLM

Evaluation & Strategy

- Benchmarks & Comparison
- Model Selection Strategy: When to pick which model
- Cost Optimization: When not to use expensive models

Module 7: Open Source & Tooling

Local Inference

- LM Studio
- Ollama

Community & Resources

- Hugging Face
- Kaggle

Observability

- LangFuse (Tracing & Monitoring)

Module 8: Coding & Frameworks

Agentic IDEs

- Antigravity, Cursor, Windsurf, OpenCode, Kiro, etc.

Backend Development

- Python with FastAPI

Agentic Frameworks

- LangChain
- LangGraph
- DeepAgents

Module 9: Design Patterns

Architectural Patterns

- Agentic RAG Designs
- Agentic AI Designs
- Memory System Designs
- Agentic HLD (High-Level Design)

Module 10: Capstone Projects (Real-World Application)

Live Projects

- 10+ Agentic Live Projects

Major Projects

- Text-to-SQL Agent: Building a "Talk to Database" system
- ClickUp Automation: End-to-end Ticketing Automation Agent

Daily Task Automation

- Identifying real-world problems
- Designing workflows to automate daily tasks using Agents